

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 20)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Describe how sulfur can be removed from coal **before** combustion.

[2 marks]

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0 1 . 2

Describe the process of dry flue gas desulfurisation and explain how the product can be used.

[3 marks]

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0 1 . 3

In wet flue gas desulfurisation the flue gases are bubbled through sodium sulfite solution. Explain **one** advantage of this method over dry FGD.

[2 marks]

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0 1 . 4

Explain how each of the following reduces NO_x emissions.

[3 marks]

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Fluidised bed combustion

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Catalytic converters

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Urea sprays

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10

0 2 . 1

Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops

☐

B Steam washing

☐

C Inflatable booms

☐

D Polymerising agents

☐

0 2 . 2

Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 2 . 3

Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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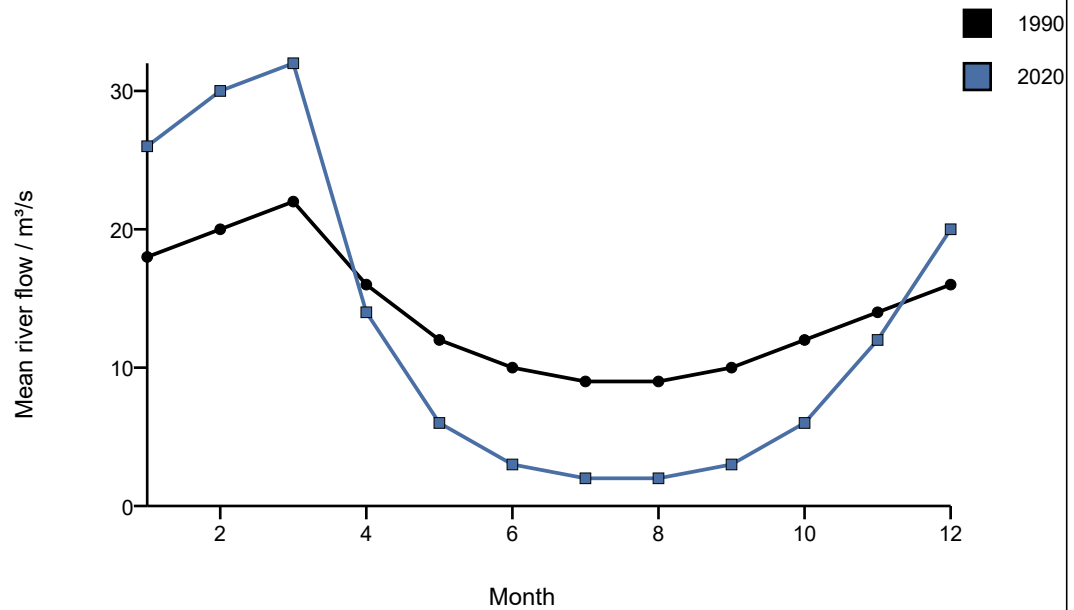
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5

0 3

Figure 1 shows the monthly flow of a river in South America that is fed partly by glacier meltwater, in 1990 and 2020. The glacier feeding the river shrank considerably between the two dates.

Figure 1



0 3

1

Compare the river flow in 1990 and 2020.

[3 marks]

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0 3

2

Explain how the shrinking of the glacier explains the differences.

[3 marks]

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03 . 3

Suggest **two** problems the change in flow causes for people living downstream.

[2 marks]

1

2

03 . 4

Explain why a glacier can be extending down its valley yet still be losing ice volume.

[2 marks]

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10

04 . 1

Describe coal gasification and coal liquefaction and explain how each increases the usefulness of coal.

[4 marks]

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04 . 2

Describe what methane hydrate is and where it is found.

[2 marks]

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04 . 3

Describe **two** proposed methods of extracting methane from hydrate deposits and explain **two** environmental risks.

[4 marks]

10

05

A rural community in Pakistan is considering a low-head helical (Archimedes screw) turbine on a river rather than joining a high-head dam scheme.

05 . 1

Explain why low-head turbines are more suitable than high-head HEP schemes for many rural areas.

[5 marks]

0 5 . 2

Suggest **one** reason why such schemes may not be installed even at suitable sites.

[1 mark]

0 5 . 3

The maximum flow rate at the site is $6.5 \text{ m}^3 \text{ s}^{-1}$ giving an output of 100 kW at 90% turbine efficiency. At 10% of maximum flow the efficiency falls to 40%. Calculate the output at 10% of maximum flow.

[2 marks]

Answer

kW

0 5 . 4

Outline **two** ways a high-head reservoir can affect the river downstream.

[2 marks]

1

2

10

0 6

A student investigated whether the number of earthworms per m^2 changes with distance from a hedge across an arable field.

0 6 . 1

Explain why systematic sampling along a transect is more appropriate than random sampling for this investigation.

[2 marks]

06 . 2

Describe how a preliminary study could be used to decide the distance between sampling sites and the number of samples at each site.

[4 marks]

06 . 3

Describe how earthworms could be extracted from a 0.5 m x 0.5 m sample area and give **two** limitations of the method.

[4 marks]

10

07 . 1

Explain why compressed air energy storage with heat recovery can achieve over 90% efficiency.

[2 marks]

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07 . 2

Explain why large-volume hot water stores lose heat more slowly per unit of stored energy than small ones.

[2 marks]

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07 . 3

Explain why molten salt rather than water is used to store heat from CSP plants.

[2 marks]

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07 . 4

Explain how high thermal mass building materials reduce energy use.

[2 marks]

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07 . 5

Describe **one** other energy storage technology.

[2 marks]

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10

08 . 1

Distinguish between **renewable** and **depletable** energy resources, giving an example of a resource that is both renewable and depletable.

[3 marks]

08 . 2

Explain why the abundance of an energy resource is not the same as the amount available for use. Give **two** examples.

[3 marks]

08 . 3

Describe the locational constraints of **two** named renewable energy resources.

[4 marks]

10

09 Table 1 shows some properties of four groups of insecticide.

Table 1

Insecticide group	Toxicity to insects	Toxicity to mammals	Persistence	Liposolubility
Organochlorines (eg DDT)	High	Low	High	High
Organophosphates (eg malathion)	High	High	Low	Low
Pyrethroids	High	Low	Low	Low
Neonicotinoids	High	Low	Moderate	Low (water soluble)

09 . 1 Use Table 1 to explain why organochlorine insecticides caused the decline of birds of prey such as the peregrine falcon in the 1950s and 1960s.

[4 marks]

09 . 2 Organophosphates replaced organochlorines for many uses. Use Table 1 to give **one** advantage and **one** disadvantage of this change.

[2 marks]

Advantage

.....

Disadvantage

.....

09 . 3

Explain why pyrethroids should not be used near rivers or fish farms.

[1 mark]

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09 . 4

Describe how the properties of neonicotinoids allow them to be used as a seed coating, and explain why this method of application reduces harm to non-target species.

[3 marks]

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10 . 1

As high-grade mineral deposits become depleted, discuss the strategies that could be used to secure future supplies of metals.

[9 marks]

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[illegible]

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1	0
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2

 State what is meant by the **stock** (resource base) of a mineral. [1 mark]

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10

Do not write outside the box

Write an essay on one of the following topics.

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Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

OR

1 1 . 2

Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

END OF QUESTIONS